Grainothèque de fleurs sauvages d'Ottawa
Wildflower Seed Library

Growing with the Land: Native Plants and Your Food Garden

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What are Native Plants?

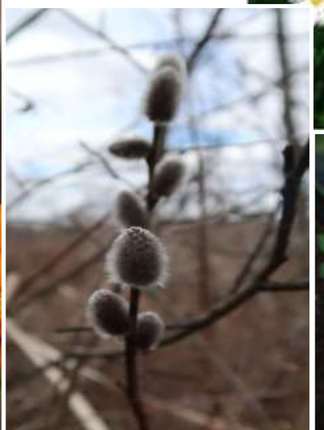
A region's soil and climate shape the plants and animals that naturally exist there.
These species are referred to as native or indigenous.
They evolved together over thousands of years, forming strong ecological connections to the other plants and animals.

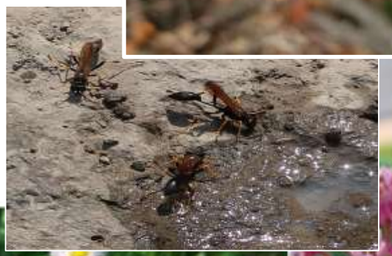













Why Incorporate Native Plants into Your Food Gardens?











They are Pollinator Powerhouses

- Native plants bring the pollinators to YOUR garden – all day, for three full seasons.
- Many fruits and vegetables have better crops when pollinated by native insects.
- Native mason bees, are much more efficient than honeybees at transferring pollen from flower to flower → more and bigger fruit production.
- Encourages and supports native pollinators and their ecosystem and reduces reliance on non-native, competing pollinators (e.g., honeybees).

- **Brassicas** (cabbage, cauliflower, broccoli, brussel sprouts, kale, turnip, collard greens, oil seeds, etc.)
  - Co-evolved with Common Eastern Bumblebee (Bombus impatiens) who can handle cooler/wetter conditions and gather the heavy, sticky pollen on their spiky hairs.
  - Honeybees come out in temps 15-25C, so may miss early and late-day blooming crops.
- **Orchard and small fruit crops** (apple, plum, cherry, raspberry, blueberry…), perform best with the “buzz pollination” of bumblebees.
- **Solanos** (Nightshade Family) (tomato, eggplant, pepper, potato) – technically self-pollinating but buzz pollination improves yield.
- **Cucurbit** (cucumbers, squashes, pumpkin, watermelon, etc.) → close blooms mid-day, need early-rising specialist (native) bees!
- …and lots more… Potatoes! Peas!…



Source: West Coast Seeds)



Poorly pollinated cucumber – there’s a solution for that!

- After *Bombus impatiens*, flower flies (e.g., hoverfly, blowfly, beefly) are the most important pollinator in agriculture (low energy needs and able to survive wide temperature range → especially good in polytunnels and greenhouses).
- Aphid-eating machines!

Spilomyia sayi, Four-lined Hornet Fly on *Anise Hyssop*
Photo: Wild Pollinator Partners



They Bring the Pest Police

- Small flowers (umbellatus-style especially) attract short-tongue bees and beneficial insects like ladybugs, hoverflies and lacewings that devour most common pests.
- In addition to beneficial insect predators, that can pick off white flies, cabbage moth grubs, aphids, etc. native plants create habitat for other friends like frogs, toads, salamanders and snakes to control unwanted garden pests.
- Minimizes using pesticides and foreign biocontrols (e.g., praying mantises) in favour of natural predators.





Zizia aurea, Golden Alexander – umbellatus form



They can Create Trap Crops

- Trap cropping is a form of organic pest management – repel unwanted pest or lure pests from your garden).
- Examples
  - Wild Bergamot (*Monarda fistulosa*) which attracts white cabbage moth – plant at least 4 feet from your crops and they will grate there.
  - Common Evening Primrose (*Oenothera biennis*) and other Oenothera plants – attract Japanese beetles away from your berries (and then you can do daily bucket drops).



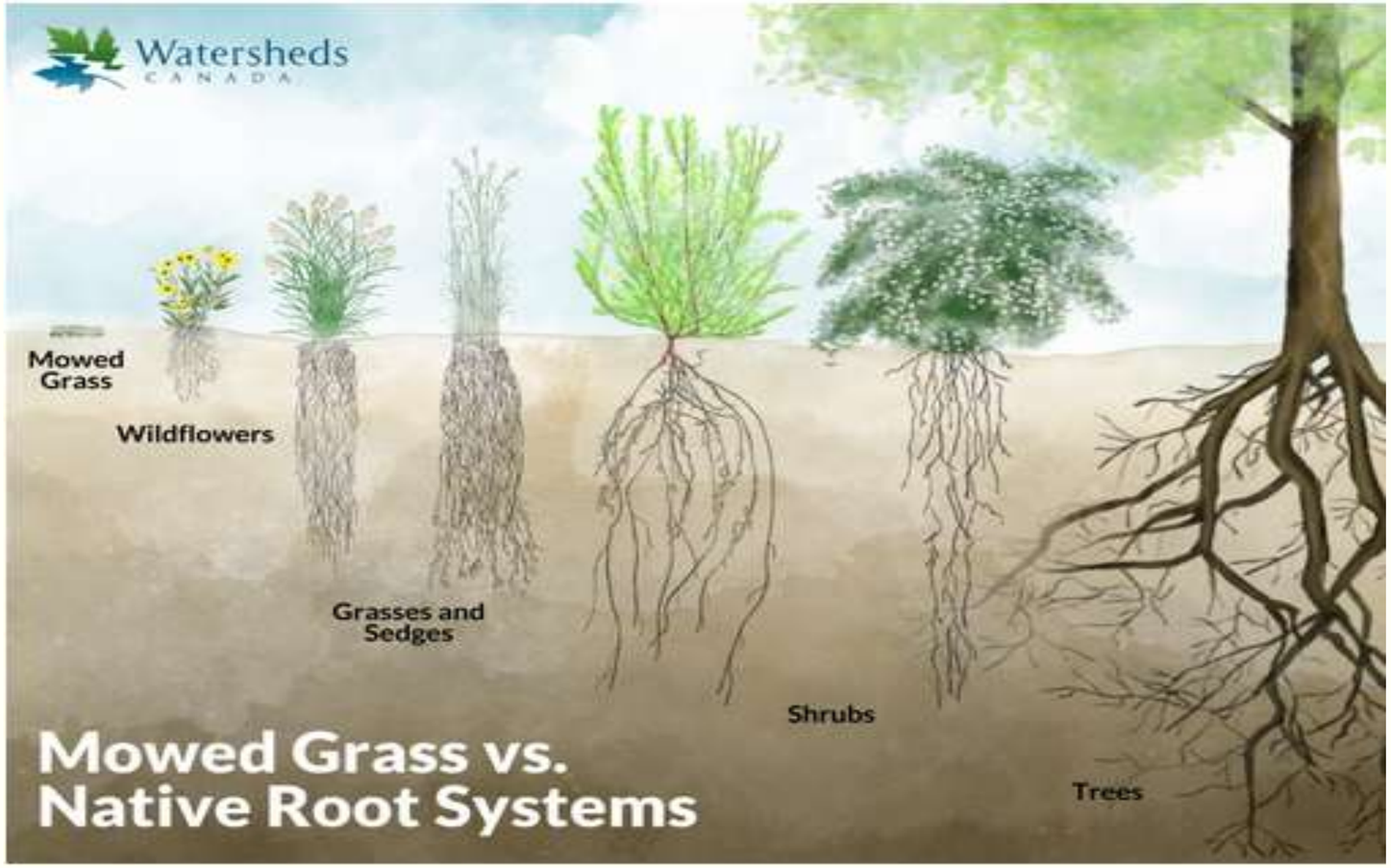


They Build Soil Health



- In addition to attracting generalist and specialist pollinators and predators, native plants provide habitat for fungi, bacteria and other microorganisms → regenerate the soil and help crops.
- Native plants are adapted to all different types of conditions; there will be suitable ones for your plot – from wetlands to desert, heavy shade to scalding sun (did you know we even have a native cactus in Ontario?)

They Mitigate the Effects of Weather and Climate Change



- Deep roots increase water retention and decrease soil erosion.
- Native cover crops act as green mulch, and provide shade to the soil, holding water, reducing weeds and invasive species, and breaking down into nutrients.
- Hedgegrows act as windbreaks/ living snow fences.

How to add Native Plants to Your Garden?











Basic Planning

- Plant pollinator patches or strips using native plants between garden beds to attract pollinators: forbs (broadleaved plants), grasses, sedges and rushes.
- Select plants that are native to your ecoregion
  - ECOREGION GUIDES: https://www.pollinator.org/guides (French & English)
  - Quebec City, Montreal, Trois-Rivières, Saint-Hyacinthe, Cornwall, Brockville, Ottawa, Gatineau, and Pembroke = St Lawrence lowlands ecoregion: https://pollinator.org/guides_canada?c=k1g Select plants that like your soil: https://sis.agr.gc.ca/cansis/publications/surveys/index.html
- Select a range of plants that will provide blooms from April to October to keep pollinators near your crops throughout the growing season.
- Choose less aggressive spreaders (by seed or rhizomes) to contain plants within their bed.
- Plants under 4’ will look tidier, and are less likely to flop into the production beds.

Implementing the Plan (I)





- Plant seeds and plugs into firm, bare soil.
- Inadequate site preparation is the primary reason native gardens fail.
  - Converting lawns, pastures, hayfields, and other areas established to non-native grasses require the most site preparation effort.
  - Herbicide and solarization are the most effective site preparation methods.
  - Site preparation methods such as occultation (covering with black plastic), repeated tillage, smother cropping, or a combination of these methods can result in a properly prepared seedbed but can take several years to be effective.

Implementing the Plan (II)

- Seeds can be mixed with inert filler such as cracked corn or non-clumping, unscented cat litter and spread by hand or broadcast by a small hand seeder.
- The seeds of many native wildflowers in areas with cold winters must go through a “cold-moist stratification” period to germinate. You can sow in the fall, winter sow in pots in the winter or CMS in the fridge for the required time.



Implementing the Plan (III)

- Plugs are most feasible in areas ¼ acre or smaller.
- Plant on 1 square foot spacing. Laying down a string or rope marked every foot, or a measuring tape can help keep spacing even
  - A small auger to dig the holes for the plugs can speed-up the planting process (and less painful).
- Plugs need watered throughout the rest of the summer until the plants enter dormancy.
- Hand weed. Can mulch but ensure it does not touch the stem of the plant → rot.
- THINK LIKE A POLLINATOR: Think like a pollinator: food, shelter, water.



Pollinator Insects' Prefer…

<u>Cross-pollination – bees, wasps, flies…</u>

Solidago canadensis, Heracleum maximum, Symphyotrichum lateriflorum var. lateriflorum Solidago juncea, Solidago nemoralis, Zizia aurea, Rudbeckia hirta, Eupatorium rugosum, Pycnanthemum tenuifolium, Symphyotrichum lateriflorum, mints

<u>Buzz pollination - Bumblebees</u> (for tomatoes, blueberries, cranberries, zucchinis, pumpkins, sweet peppers, potatoes, eggplant)

Eupatorium perfoliatum, Pycnanthemum tenuifolium, Symphyotrichum novae-angliae, Echinacea purpurea, Veronicastrum virginianum



Osmia lignaria, Blue Orchard Mason Bee



Blue Orchard Mason Bee info: https://www.natureconservancy.ca/en/what-we-do/resource-centre/featured-species/insects-and-spiders/blue-orchard-mason-bee.html

Predatory Insects' Prefer…

Assassin Bugs (*Reduvidae*)
Eat aphids, beetles, true bugs, caterpillars, flea beetles, grasshoppers, leaf hoppers
Top native plants: *Symphyotrichum ericoides, Helenium autumnale, Rudbeckia hirta,* all goldenrods

Ambush Bugs (*Phymatidae*)
Eat aphids, beetles, true bugs, caterpillars, flea beetles, grasshoppers, leaf hoppers
Top native plants: *Euphorbia corollata, Eupatorium perfoliatum*

Predatory Stink Bugs (*Pentamoidae*)
Eat aphids, immature beetles, true bugs, immature flea beetles, Colorado Potato Beetle eggs
Top native plants: *Euthamia graminifolia, Helenium autumnale*

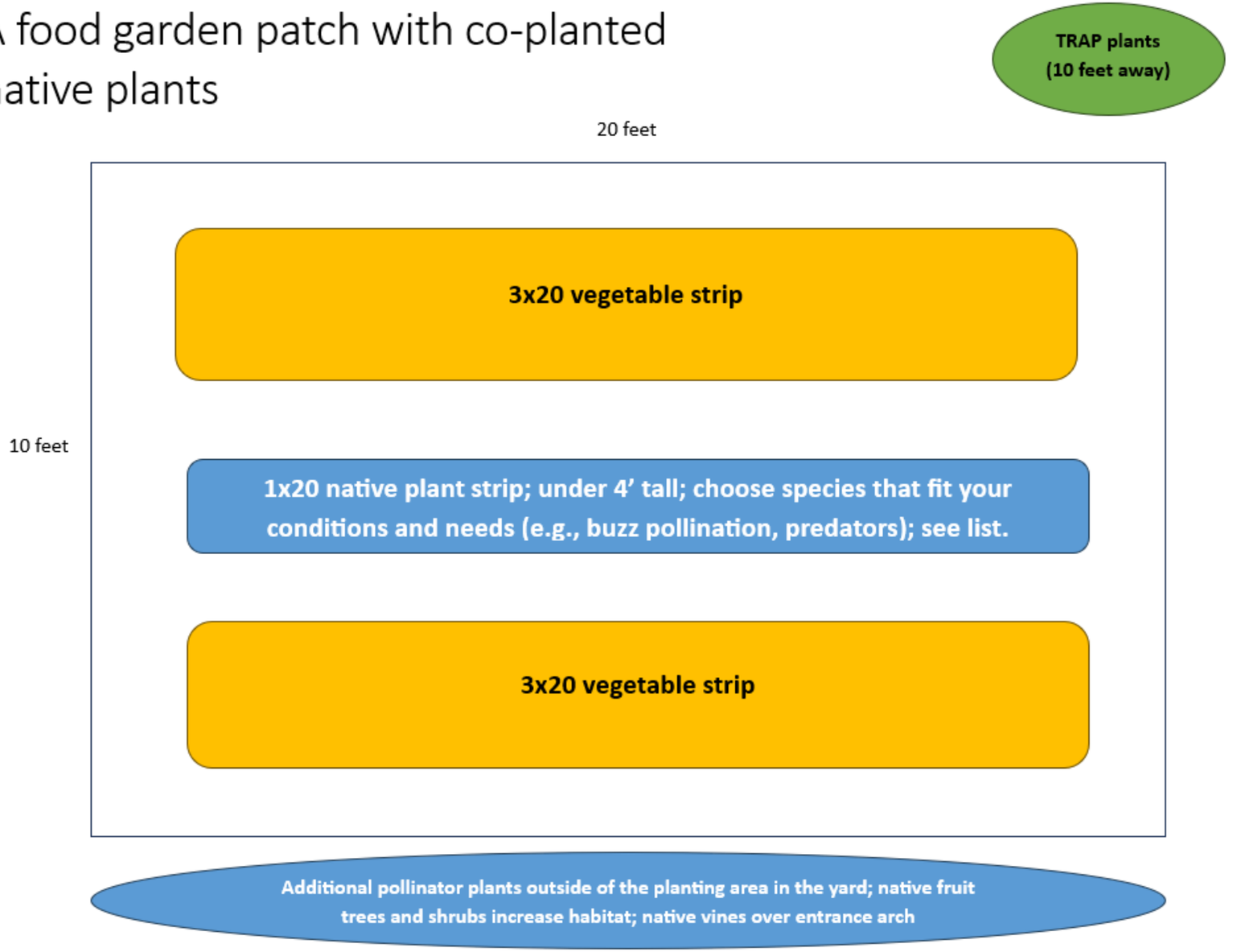
Damsel Bugs (*Nabidae*)
Eat leaf beetles, small caterpillars, leaf beetles, leaf hoppers, mites, thrips, insect eggs, aphids
Top native plants: All goldenrods *(Solidago), Pycnanthemum tenuifolium, Amorpha canescens, Asclepias syriaca, Coreopsis tripteris Dalea candida*

Big Eyed Bugs (*Geocoridae*)
Eat leaf beetles, small caterpillars, leaf beetles, leaf hoppers, mites, thrips, insect eggs, aphids
Top native plants: All goldenrods (*Solidago*)! *Erigeron annuus, Coreopsis tripteris, Amorpha canescens*

Predatory Mites (*Phytoselids*)
Eat small caterpillars, mites, immature thrips, insect eggs, whiteflies
Top native plants: *Erigeron annuus, Helenium autumnale, Solidago rigida, Monarda punctata, Verbena hastata*

Predatory/Parasitoid Wasps (*Brachanoid, Vespids, Sphecids, Chalcid and Ichneumon*)
Eat caterpillars (tomato hornworms), mites, flea beetles, leaf beetles, insect eggs, white flies…
Top native plants: *Rudbeckia lainata, Heracleum maximum, Pycnanthemum tenuifolium, Zizia aurea, Pycnanthemum virginianum, Symphyotrichum lateriflorum, Solidago juncea, Eupatorium perfoliatum, Verbesina alternifolia, Agastache Foeniculum, Eurybia macrophylla, Fragaria virginiana, Taenidia integerrima*

A food garden patch with co-planted native plants



Native Plants also Build Patience ☺



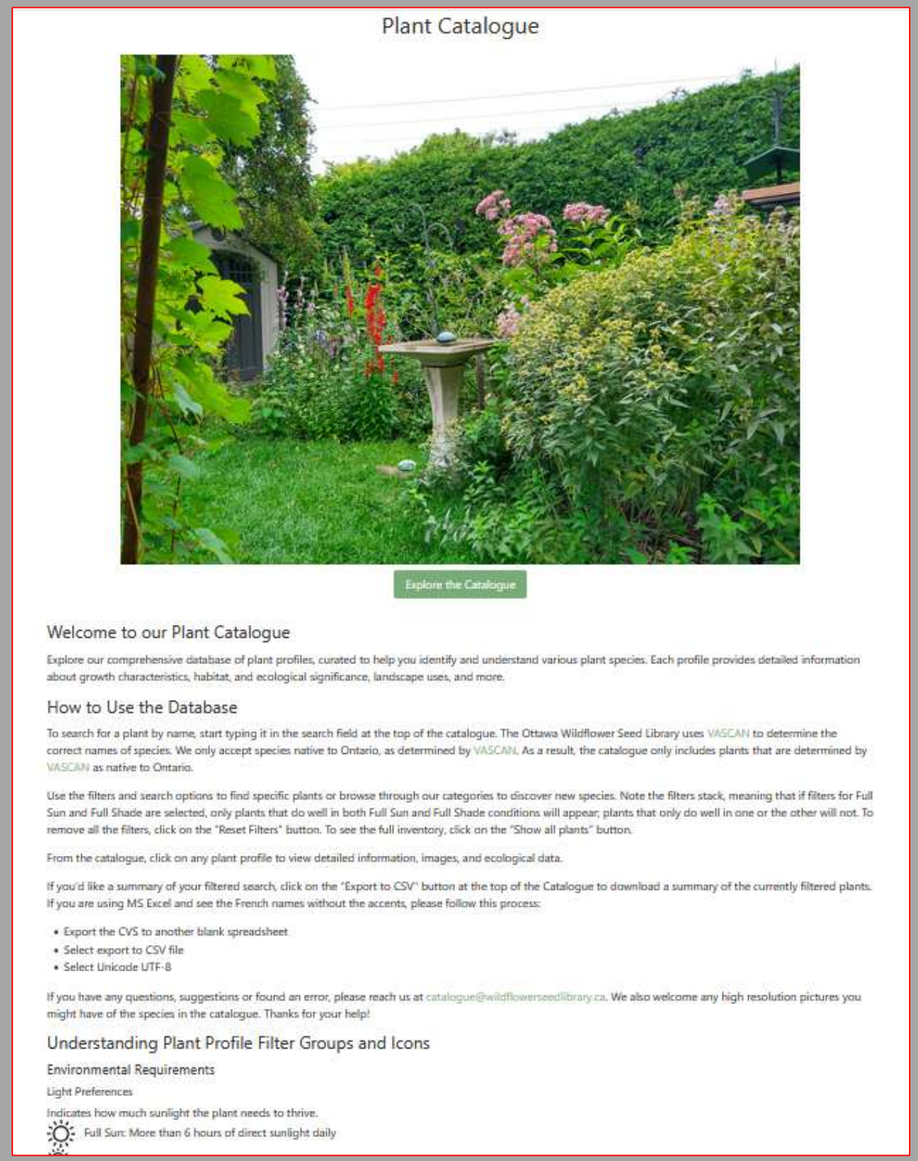
- Native plants take 3 years or more to reach their full size. Most spend Y1 putting down roots.
- Native, perennial pollinator habitat of any shape and size planted in or near your garden will improve pollination and decrease populations of garden pests.

OWSL Catalogue

Thanks for joining me tonight! Check out the resources on our website, especially the new catalogue:

https://catalogue.wildflowerseedlibrary.ca

Search by selection criteria such as light, soil, or purpose to native plant options for your garden!



Reference & Resources







- Janice Keil North America Native Plant Society (NANPS), “Rewilding the Veggie Garden”: https://www.youtube.com/watch?v=uTh9oF0CBXg
- Pollinator Partnership (choose Canadian resources from the site): https://www.pollinator.org/
- Michigan State University, “Native Plants and Ecosystem Services: Uses for Agriculture”: https://www.canr.msu.edu/nativeplants/getting_started/agriculture
- Grow Native. “Native plants for livestock pastures” (US): https://grownative.org/learn/native-for-farms/natives-for-livestock/
- Northern Wildflowers: https://northernwildflowers.ca/blogs/our-blog/companion-planting-for-increased-garden-success
- Articles on edible native plant crops: https://lovenativeplants.com/npa, and In Our Nature https://www.inournature.ca/ediblenativeplants and Native Plant Connection Ontario https://nativeplantconnection.com/edible-ontario-native-plants/
- Agriculture and Agri-food Canada, “Detailed soil surveys in Canada”: https://sis.agr.gc.ca/cansis/publications/surveys/index.html
- The eco-transformation of a farm: https://haliburtonfarm.org/biodiversity/
- Xerces Society “Reducing Risks to Wildlife from Insecticide-Treated Seed”: https://www.xerces.org/publications/fact-sheets/reducing-risks-to-wildlife-from-insecticide-treated-seed
- National Farmers’ Union - Native Seed Sourcing for Restoration & Farms https://nfuontario.ca/op-eds/planting-the-seed-growing-native-plants/
- Wasp and pollinator fact sheets: https://www.pollinatorsnativeplants.com/resources.html
- Seeds of Diversity, information sheets on pollinators: https://seeds.ca/pollinator/bestpractices/images/PDF_Wild%20pollinators.pdf



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